Supervised Learning

$$X \xrightarrow{f} Y$$

goal: Derive f

1. Collect data from x, and corresponding Y

training data: $X \xrightarrow{f_{data}} Y$

assume $f_{est} \sim f_{data}$

$f_{est}(\theta, \theta_0) \Rightarrow$ find θ and θ_0

$\Rightarrow$ minimize $J = Loss(f_{est}, f_{data})$

$= Loss(\theta, \theta_0, X_{data}, Y_{data}) + \alpha R(\theta)$

☆ $f_{data} \neq f$

Hyperparameter: α (must choose)

α is not determined by the min of J.

SVM : Max the margin of decision

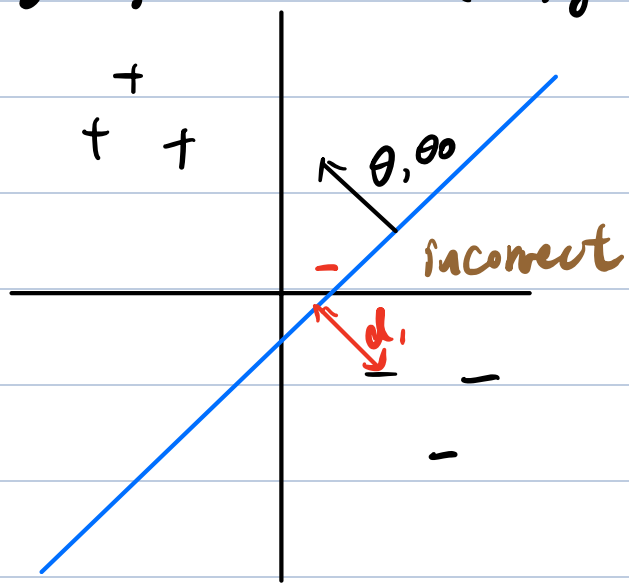
margin :

$D : \{(x^i, y^i)\}, \; i = 1 \ldots n$

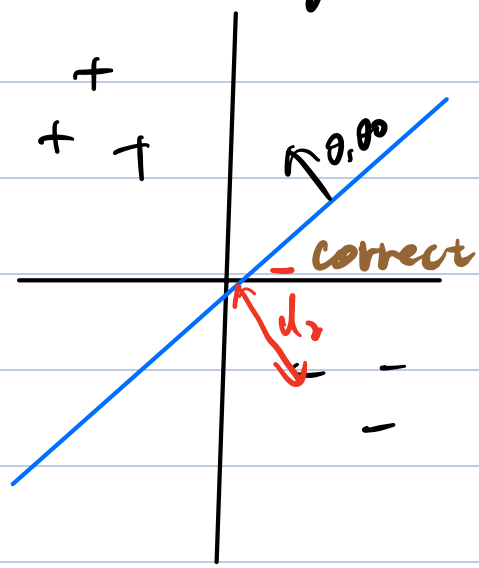
dist from point i to decision boundary (γ)

$$\gamma = \frac{y^i(\theta \cdot x^i + \theta_0)}{|\theta|}$$

margin, $d = \min(x^i, y^i \in D)$ of $\gamma(x^i, y^i, \theta, \theta_0)$



Less margin



Larger margin

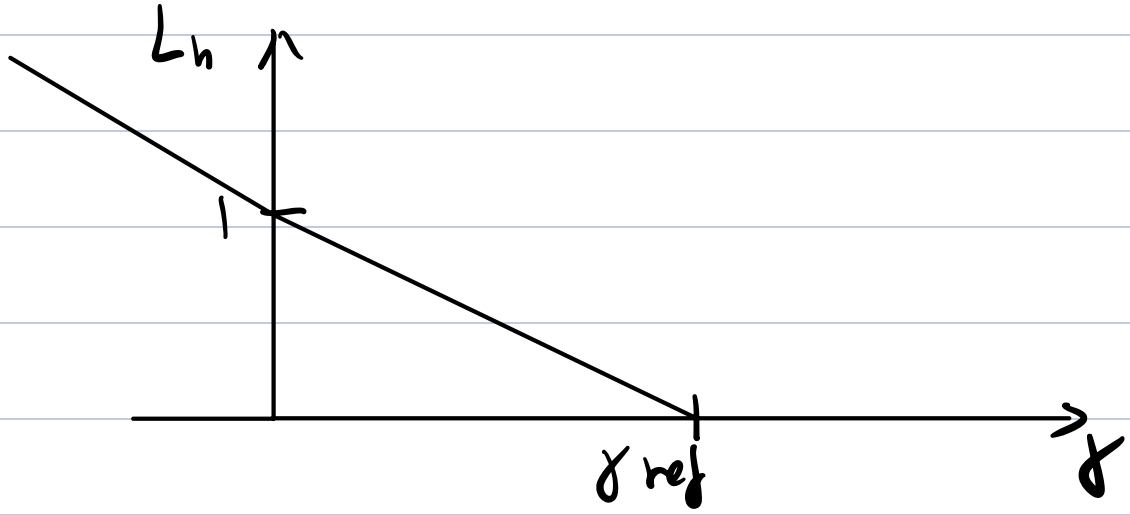
In SVM we want to maximize the margin

=> large d -> more generalizable.

In SVM

Loss = hinge Loss

$$L_h = f\left(\frac{\gamma}{\gamma_{ref}}\right) = \begin{cases} 1 - \frac{\gamma}{\gamma_{ref}} & \gamma < \gamma_{ref} \\ 0 & \text{, otherwise} \end{cases}$$



γ_{ref} : margin of decision boundary

Regularization Term

→ more generalizable

$\Rightarrow$ max γ_{ref} $\Rightarrow$ min $\frac{1}{\gamma_{ref}}$ $\Rightarrow$ min $\frac{1}{\gamma_{ref}^2}$

For SVM

$$J = \frac{1}{n}\sum_{i=1}^{n} L_h\left(\frac{\gamma}{\gamma_{ref}}\right) + \alpha\left(\frac{1}{\gamma_{ref}^2}\right)$$

How to define γ_{ref}

We can scale θ by any const $(c \cdot \theta)$
(not changing decision boundary)

$\therefore$ we can scale θ

$$\Rightarrow y^m (\theta \cdot x^m + \theta_0) = 1$$

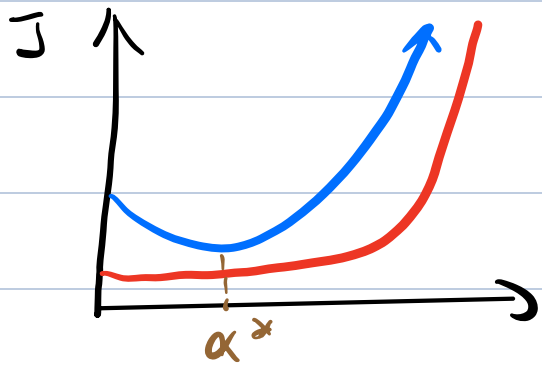
$$\gamma_{ref} = \frac{1}{|\theta|}$$

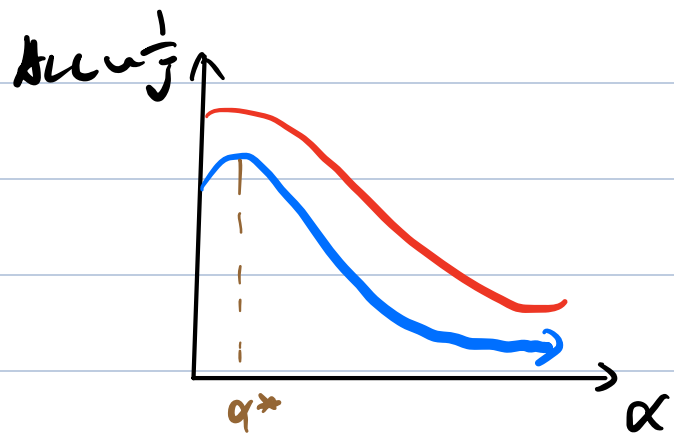
$$J = \frac{1}{n} \sum_{i=1}^{n} L_n(y^i(\theta \cdot x^i + \theta_0)) + \alpha |\theta|^2$$

How α affects model

Training

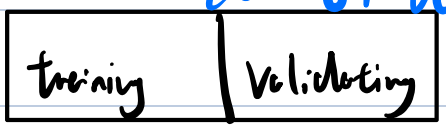
Testing





Cross Validation

(For find α)



Training data

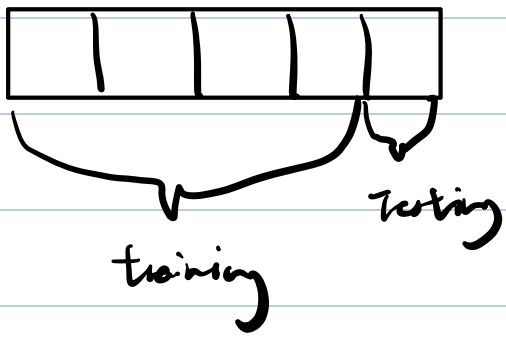
$S(\alpha)$: 代表准确率

parameter N : how many time to segment

e.g. $n = 5$

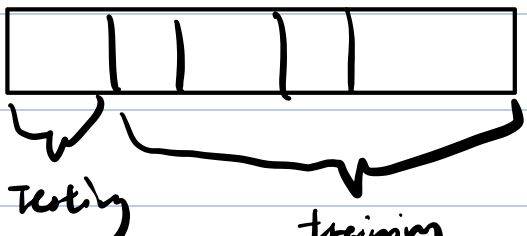
$\alpha : \alpha_1 \dots \alpha_n$, $\alpha_1 < \alpha_1 < \dots < \alpha_n$

$\alpha = \alpha_1$



iter1 $\quad acc = S_1(\alpha_1)$

iter2



$acc = S_2(\alpha_1)$

repeat 5 Times

$S_5(a_1)$

$$S(a_1) = \frac{1}{n}\sum_{i=1}^{n} S_n(a_1)$$

Repeat for each a in a_1 to a_n

Find max $S(a^*)$ a^* is the best